

# Temple Management System

## Business Logic & Calculation Guide

A single-source explanation of every calculation, business rule and workflow used by the application.  
Read this before doing any audit, migration, or hand-off to a new developer.

| Section | Topic                                                       |
|---------|-------------------------------------------------------------|
| 1       | Penalty Engine (Subscription / Death / Festival / Marriage) |
| 2       | Chit Fund - Core Accounting Identity                        |
| 3       | Chit Fund - Share Amount vs Collected Share Amount          |
| 4       | Chit Fund - Settlement Application Flow                     |
| 5       | Interest Module - EMI vs Regular Interest                   |
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# 1. Penalty Engine

The penalty engine applies a **flat monthly late-fee** to every member whose payment is overdue. It is deterministic, idempotent (safe to re-run) and works uniformly across the four contribution modules where members can owe money.

## 1.1 The rule

**Penalty = £ 25 × (number of full calendar months elapsed since the due date, no grace period)**

## 1.2 How "missed months" is counted

We use the calendar-month difference between the due date and today. A month is only counted once the same day-of-month has passed. So:

| Due date   | Today      | Missed months            | Penalty |
|------------|------------|--------------------------|---------|
| 2024-04-10 | 2024-04-30 | 0                        | £0      |
| 2024-04-10 | 2024-05-09 | 0 (not yet a full month) | £0      |
| 2024-04-10 | 2024-05-10 | 1 (exactly one month)    | £25     |
| 2024-04-10 | 2024-05-11 | 1                        | £25     |
| 2024-04-10 | 2024-07-15 | 3                        | £75     |
| 2024-12-10 | 2025-03-10 | 3 (spans year boundary)  | £75     |
| 2024-04-08 | 2026-06-30 | 26                       | £650    |

## 1.3 Where the due date comes from

Each row in `PeoplesAmountDetails` is linked to one of four items. The due date is pulled from the linked item:

|                              |                                                                            |
|------------------------------|----------------------------------------------------------------------------|
| <b>Subscription Tariff</b>   | <code>sub_tariff.to_date</code>                                            |
| <b>Festival contribution</b> | <code>festival.penalty_start_date</code> or <code>festival.end_date</code> |
| <b>Death tariff</b>          | <code>death.penalty_apply_date</code> or <code>death.date</code>           |
| <b>Marriage tariff</b>       | <code>marriage.marriage_date</code>                                        |

## 1.4 Idempotency — why re-running is safe

Every call to the engine **overwrites** `penalty_amount` with the freshly computed value; it never adds on top of the old value. The already-paid portion of the penalty is preserved separately in `penalty_balance`:

**`penalty_amount = £ 25 × missed_months`**  
**`penalty_balance = penalty_amount – penalty_already_paid`**

Running the engine 100 times a day gives exactly the same result as running it once. This is what makes the "Recompute" button on the Pending Penalty List page safe.

## 1.5 API endpoints

|                                     |                                             |
|-------------------------------------|---------------------------------------------|
| <b>GET /api/penalty/pending/</b>    | List of members with pending penalty        |
| <b>POST /api/penalty/recompute/</b> | Force full recompute; returns summary       |
| <b>GET /api/penalty/summary/</b>    | Totals only (rate, scanned, updated, total) |

## 2. Chit Fund - Core Accounting Identity

A chit fund is essentially a rotating investment pool. The system keeps four running totals on the ChitFundsDetails row, and they must always satisfy one identity:

$$\text{Cash\_In\_Hand} = \text{Pool} + \text{Profit} + \text{Collected\_Principal} - \text{Principal\_Given}$$

If this identity is ever violated, the books are out of sync and a fix must be applied before more transactions are booked.

### 2.1 What each total means

|                                   |                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------|
| <b>Pool</b>                       | Total money put in by all investors + management. Set once at chit creation.  |
| <b>Principal Given Amount</b>     | Cumulative principal money loaned out to borrowers over the life of the chit. |
| <b>Collected Principal Amount</b> | Cumulative principal repaid by borrowers so far.                              |
| <b>Profit Amount</b>              | Cumulative interest + penalty income collected from borrowers so far.         |
| <b>Cash In Hand Amount</b>        | Money physically remaining in the chit's account today.                       |

### 2.2 Worked example - Chit Fund #1 (AMMAN FINANCE)

|                                   |                               |
|-----------------------------------|-------------------------------|
| <b>Management Invested Amount</b> | &#377; 10,000 (1 share)       |
| <b>Outer Invest Amount</b>        | &#377; 14,00,000 (140 shares) |
| <b>Fixed Share Amount</b>         | &#377; 10,000                 |
| <b>Total Share Count</b>          | 141                           |
| <b>Pool</b>                       | &#377; 14,10,000              |
| <b>Principal Given Amount</b>     | &#377; 1,04,48,500            |
| <b>Collected Principal Amount</b> | &#377; 87,88,450              |
| <b>Profit Amount</b>              | &#377; 9,76,050               |
| <b>Cash In Hand Amount</b>        | &#377; 7,26,000               |

Let us verify the identity:

$$\begin{aligned} &\text{Pool} + \text{Profit} + \text{Collected} - \text{Given} \\ &= 14,10,000 + 9,76,050 + 87,88,450 - 1,04,48,500 \\ &= 7,26,000 \quad \text{--> matches Cash In Hand} \quad \text{OK} \end{aligned}$$

### 2.3 What changes each total, and when

- **Loan given** (create interest with type = Chit Fund Interest):  
cash\_inhand -= principal\_amt  
principal\_given += principal\_amt
- **Loan payment** (collection accept, non-installment):  
collected\_principal += amount  
cash\_inhand += amount + interest\_amount + penalty\_amount

```
profit_amount += interst_amount + penalty_amount
```

- **Loan EMI payment** (Installment Interest):

```
new_principal = amount - interest_portion  
collected_principal += new_principal  
cash_inhand += amount + penalty (after bugfix)  
profit_amount += interest_portion + penalty
```

- **Loan edit** (interest PUT):

Reverse OLD principal, then apply NEW principal (*after bugfix*).

### 3. Chit Fund - Share Amount vs Collected Share Amount

Every investor has two fields that look similar but mean very different things. Confusing them was the root cause of the "Share Amount is always 0" issue you reported.

#### 3.1 Field definitions

|                                |                                                                            |
|--------------------------------|----------------------------------------------------------------------------|
| <b>collected_share_amount</b>  | The LIVE running total of the investor's share of profit. Updated on EVERY |
| <b>share_amount</b>            | The FINAL settled share (only set at settlement time). Stays 0 while the c |
| <b>final_settlement_amount</b> | The total pay-out to the investor at settlement = investment_amt + share   |

#### 3.2 How collected\_share\_amount grows

On every borrower payment the interest and penalty portion is distributed among all investors proportional to their share count. Management takes its cut first, the rest is split.

```
temple_amt = profit_amt × set_profit_percent / 100
remaining = profit_amt – temple_amt
per_share = remaining / total_share_count
For every investor: collected_share_amount += share_count × per_share
```

#### 3.3 Worked example - Investor N. SUNDAR (Chit #1)

|                                |                                                    |
|--------------------------------|----------------------------------------------------|
| <b>investment_amt</b>          | &#8377; 5,00,000                                   |
| <b>share_count</b>             | 50                                                 |
| <b>collected_share_amount</b>  | &#8377; 3,13,960.00 (accumulated over many months) |
| <b>share_amount</b>            | 0.00 (chit not yet settled)                        |
| <b>final_settlement_amount</b> | 0.00 (no settlement application yet)               |

**Note:** The old UI displayed share\_amount in the Chit Fund View which is always 0 for active chits. That is why every member showed 0. Now the UI displays collected\_share\_amount, which is the true LIVE value each investor has earned so far.

## 4. Chit Fund - Settlement Application Flow

A settlement application is a formal record that an investor is ready to receive their pay-out. The flow has three stages, each leaving a different visible mark in the system.

### 4.1 Stage A - Before submission (default)

The Chit Fund View shows the investor's live numbers but the Final Settlement Amount row is **hidden** because the value is 0. This is by design: nothing to settle yet.

### 4.2 Stage B - Filling the Settlement Application form

In `/chitfund_settlement_application` the user picks a chit fund, then picks the investor. Four read-only fields auto-populate:

|                        |                                                 |
|------------------------|-------------------------------------------------|
| <b>Invested Amount</b> | investment_amt from the investor                |
| <b>Share Count</b>     | share_count from the investor                   |
| <b>Share Amount</b>    | collected_share_amount from the investor        |
| <b>Total Amount</b>    | Invested + Share Amount (what will be paid out) |

### 4.3 Stage C - On submission

The `add_chit_fund_settlement_application_details` endpoint runs the following on the linked investor row:

```
investor.share_amount      = collected_share_amount    # freeze the value
investor.application_date   = settlement_date
investor.action             = False                    # locked, cannot be edited
investor.final_settlement_amount = investment_amt + share_amount
```

After this, the Chit Fund View instantly shows a new row for that member:

**Final Settlement Amount : ■ XXX,XXX.XX (bold green)**

### 4.4 What each stage looks like for N. SUNDAR

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| <b>Before submission</b> | Final Settlement Amount row - hidden              |
| <b>After submission</b>  | Final Settlement Amount : ₹ 8,13,960.00           |
| <b>Formula</b>           | 5,00,000 (invested) + 3,13,960 (share) = 8,13,960 |

## 5. Interest Module - EMI vs Regular Interest

The interest module supports two loan styles. Understanding the difference is critical because the money-flow formulas differ.

### 5.1 Regular interest (interest-only)

Borrower pays interest each period; principal is returned in one lump sum later. Each period payment is split explicitly:

```
temp_family.amount = principal part (usually 0 mid-loan)
temp_family.interst_amount = interest for the period
temp_family.penalty_amount = late fee if any
```

### 5.2 EMI / Installment interest

Borrower pays a fixed EMI each month; principal and interest are baked into a single number:

```
temp_family.amount = full EMI (principal + interest)
new_pro_amount = interest portion (computed by rate & period)
new_principal_amt = amount - new_pro_amount (principal portion)
```

***Note:** The old code added `interst_amount` to `cash_inhand` for EMI payments **on top of** the already-inclusive EMI, double-counting the interest. This has been fixed.*

### 5.3 Chit-fund penalty-in-arrears rule

The chit-fund interest module also charges a penalty when a borrower is late. It applies once a full calendar month has passed since the last `interest_date`. In the original code this was gated by:

```
# Old, buggy:
if interest_date.year == today.year and interest_date.month != today.month:
    apply_penalty()
```

That silently skipped every loan that had been created in a previous calendar year. Fixed to a year-agnostic condition:

```
# New, correct:
if interest_date + relativedelta(months=1) <= today:
    apply_penalty()
```



## 6. Bugs Fixed in This Session

### 1. Interest year condition

**Where:** interest/views.py line 178

**Before:** Penalty skipped whenever the loan was created in a previous year.

**After:** Replaced year+month check with a year-agnostic interest\_date + 1 month <= today check.

### 2. EMI cash-in-hand double-count

**Where:** collection/views.py line 624-632

**Before:** For Installment Interest, cash\_inhand was incremented by both amount (full EMI) AND interst\_amount, double-counting the interest.

**After:** Now adds only amount + penalty\_amount . Profit correctly gets new\_pro\_amount + penalty .

### 3. Chit-fund loan edit not re-applied

**Where:** interest/views.py line 413-419

**Before:** When editing an existing chit-fund loan the OLD principal was reversed but the NEW principal was never re-applied, leaving cash\_inhand and principal\_given out of sync.

**After:** Now: reverse old, then apply new principal in one atomic block.

### 4. Share Amount showing 0 for all members

**Where:** ChitFundListView.jsx display

**Before:** UI read share\_amount which is only set at settlement; every active-chit member showed 0.

**After:** UI now reads collected\_share\_amount ?? share\_amount ?? 0 .

### 5. Final Settlement Amount showing 0 for everyone

**Where:** ChitFundListView.jsx display

**Before:** Field always displayed even when no settlement application was submitted, causing confusion.

**After:** Row is now hidden when final\_settlement\_amount == 0; appears only after settlement submission.

### 6. Sidebar overlap on all pages

**Where:** layout/Partials/Style.js and DashboardLayout.jsx

**Before:** SideMenuLayout was position: fixed but the right had no margin-left offset, so header text and home image rendered under the sidebar.

**After:** Added ContentLayout with margin-left: 280px (80 collapsed, 0 on mobile) and z-index on sidebar.

## 7. Member List images broken

**Where:** `temple_proj/settings/settings.py` and `urls.py`

**Before:** `MEDIA_URL` was `/media/` but K8s ingress only routes `/api/*` to backend, so image URLs 404-ed. `DEBUG=False` also disabled Django `static()` serving.

**After:** Changed `MEDIA_URL` to `/api/media/` and added an explicit `re_path(r'^api/media/...', serve, ...)` route. Added `onError` fallback on Member List images.

## 8. Penalty engine did not exist

**Where:** `amount/penalty_engine.py` (new file)

**Before:** Each module set a static one-time penalty; no accumulation per missed month; `PendingPenaltyList` screen was hardcoded mock data.

**After:** New idempotent engine + REST endpoints + real UI. Applies ■ 25 x missed\_months across all four contribution modules.

## 9. Settlement Application form missing figures

**Where:** `ChitFundSettlementAppln.jsx`

**Before:** The form didn't show Invested / Share Count / Share Amount / Total Amount when picking an investor.

**After:** Added 4 auto-populating read-only fields; `Total Amount = Invested + collected_share_amount`.

## 10. Settlement Application list missing figures

**Where:** `chit_fund/serializers.py` and list view

**Before:** List showed only application no + investor name.

**After:** Added derived Serializer Method Fields (`share_count`, `share_amount`, `investment_amt`, `total_amount`) pulled from the linked investor; added three new columns to the list & print tables.

## 11. Duplicate penalty rows on Balance Sheet ledger

**Where:** `my_tasks/views.py` penalty scheduled task

**Before:** For every re-run of the task, a new `TempleMemberReport` row was inserted for the same (member, tariff/festival/death). Also `amount_balance` and `total_bal_amt` on `PeoplesAmountDetails` were incremented by `penalty_amount` every run, inflating the running balance.

**After:** Added two idempotency guards: (a) only add penalty to running balances if penalty flag is not already True; (b) skip `TempleMemberReport` insert if a row with the same (member, tariff/festival/death, type\_choice) already exists. Also added `/app/backend/cleanup_penalty_duplicates.py` to remove existing duplicates and recompute `balance_amt` column.

## 7. Data Model Cheat-Sheet

The main tables and their key money-fields. Keep this next to you when reading any calculation code.

### PeoplesAmountDetails (amount)

|                        |                                                   |
|------------------------|---------------------------------------------------|
| <b>amount</b>          | Base amount owed for a contribution period        |
| <b>total_paid_amt</b>  | How much of the base has been paid                |
| <b>total_bal_amt</b>   | Outstanding base amount = amount - total_paid_amt |
| <b>penalty</b>         | Boolean flag: any pending penalty?                |
| <b>penalty_amount</b>  | Total penalty computed by the engine              |
| <b>penalty_balance</b> | Unpaid portion of penalty_amount                  |
| <b>paid</b>            | True once base + penalty is fully cleared         |

### ChitFundsDetails (chit\_fund)

|                                   |                                             |
|-----------------------------------|---------------------------------------------|
| <b>management_amt</b>             | Cash the temple/management itself puts in   |
| <b>outer_invest_amount</b>        | Cash all investors put in                   |
| <b>fixed_share_amt</b>            | Money value of one share                    |
| <b>management_share_count</b>     | How many shares management holds            |
| <b>investors_share_count</b>      | Sum of shares held by all outside investors |
| <b>total_share_count</b>          | management + investors                      |
| <b>set_profit_percent</b>         | % of profit that goes to management         |
| <b>principal_given_amount</b>     | Total loans given out to date               |
| <b>collected_principal_amount</b> | Total principal repaid to date              |
| <b>profit_amount</b>              | Total interest + penalty collected to date  |
| <b>cash_inhand_amount</b>         | Money currently in the chit's bank          |

### ChitFundInvesters (chit\_fund)

|                                |                                               |
|--------------------------------|-----------------------------------------------|
| <b>investor_name</b>           | Human-readable investor name                  |
| <b>investment_amt</b>          | Principal put in by this investor             |
| <b>share_count</b>             | How many shares this investor holds           |
| <b>collected_share_amount</b>  | LIVE profit share earned so far (grows)       |
| <b>share_amount</b>            | FROZEN share on settlement (0 until then)     |
| <b>application_date</b>        | Date the settlement application was submitted |
| <b>final_settlement_amount</b> | investment + share_amount (set at settlement) |

|               |                                                           |
|---------------|-----------------------------------------------------------|
| <b>action</b> | Editable? False after settlement application is submitted |
|---------------|-----------------------------------------------------------|

### ChitFundsettleApplication (chit\_fund)

|                                 |                                   |
|---------------------------------|-----------------------------------|
| <b>settlement_aplication_no</b> | Auto-generated: CHIT-APP<n>       |
| <b>chitt_fund</b>               | FK to ChitFundsDetails            |
| <b>investers</b>                | FK to ChitFundInvesters           |
| <b>settlement_date</b>          | Date entered by staff on the form |
| <b>comments</b>                 | Optional free-text notes          |

*End of guide. This document reflects the code state as of the current session. Keep the PDF alongside the codebase; regenerate whenever a core formula changes.*